

DANIEL NG

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Education

University of British Columbia

BASc in Electrical Engineering (Co-op, available May 2026)

Expected Graduation: May 2028

Vancouver, BC

Experience

Norsat International Inc.

June 2025 – August 2025

Hardware Intern

Richmond, BC

- Characterized Ku-Band Block-Up Converters (gain, P_{sat} , P1dB, spurious, IMD3, noise figure) using VNAs, signal analyzers, oscilloscopes, and multimeters to ensure 100% compliance with hardware specifications.
- Optimized RF power module stability by precisely sweeping DAC control codes, significantly reducing variance in output power prior to final acceptance testing.
- Assembled and troubleshoot BUCs and LNBs at the PCBA level, performing microscopic SMT and THT soldering and inspection to maintain quality standards for high-sensitivity MMIC chips.
- Verified and troubleshoot SSPA boards by conducting continuity tests and cross-referencing schematics to identify and rectify DC bias instability, ensuring 100% yield for high-power MMIC stages.

UBC Formula Design Team

September 2025 – Present

Firmware Subteam: Timing Gates Project Lead

Vancouver, BC

🐙 [GitHub Repository](#)

- Led end-to-end development of a laser receiver, including optomechanical shrouding coupled with a custom 3.3 V CMOS analog front end and 2.4 GHz communication between master-slave ESP32-S3 MCUs to calculate speed and lap times.
- Characterized baseline photodiode thermal dark current and TIA voltage noise to calibrate an adjustable hysteresis threshold via potentiometer, targeting a 20 dB SNR and BER $< 10^{-7}$ to ensure high-speed data integrity.

Projects

Interrupt-Driven Alarm Clock | *A51 Assembly, N76E003 (8051) MCU*

January 2026

🐙 [GitHub Repository](#)

- Engineered a real-time settable clock using a 16.6 MHz oscillator to drive timers and interrupts.
- Utilized Timer 0 to generate a stable 2 kHz square wave for audio signaling and Timer 2 to manage high-precision half-second BCD clock increments.
- Interfaced a 4-bit LCD for time/alarm display and implemented a sampling-based debouncing algorithm coupled with decoupling capacitors for pushbutton inputs to eliminate noise.

FPGA VGA Driver | *C, Altera DE10-Lite FPGA, MMIO, VGA*

November 2025

🐙 [GitHub Repository](#)

- Developed a custom VGA graphics driver using MMIO to manage frame buffers, rendering dynamic light traces for a fully functional, one-player "Tron Light Cycle" game, while maintaining real-time pixel updates at VGA timing constraints.
- Implemented interrupt-driven input handling to handle asynchronous pushbutton events, reducing input latency compared to traditional polling.
- Engineered an autonomous robot opponent with predictive collision logic, performing real-time memory reads of the VGA buffer to scan for obstacles and update movement vectors.

RISC-V Simple CPU | *SystemVerilog, Altera DE10-Lite FPGA, Quartus Prime, Questa*

November 2025

🐙 [GitHub Repository](#)

- Architected and implemented a single-cycle RISC-V processor in SystemVerilog with MMIO on Altera's DE10-Lite with functioning register file, ALU, and control unit, supporting a 10-instruction subset of the RISC-V ISA.
- Wrote a self-checking non-synthesizable testbench with assertion-based verification to verify instruction-level correctness and used Questa to trace signals through the synchronous timing datapath.

Technical Skills

Digital Design/HDL/Low-Level: SystemVerilog, RISC-V Assembly, A51 Assembly, FSM Implementation & Design

Embedded Systems: CAN, USB, SPI, UART, I2C, ESP32 and 8051-family MCUs, C Embedded Firmware, RV32 Bare Metal, MMIO

Design Tools: Quartus Prime, Questa/ModelSim, Altium Designer, CPUlator

Hardware & Lab: VNA, Signal Analyzer, PCB Design and Rework, SMT/THT Soldering, Oscilloscope, Multimeter